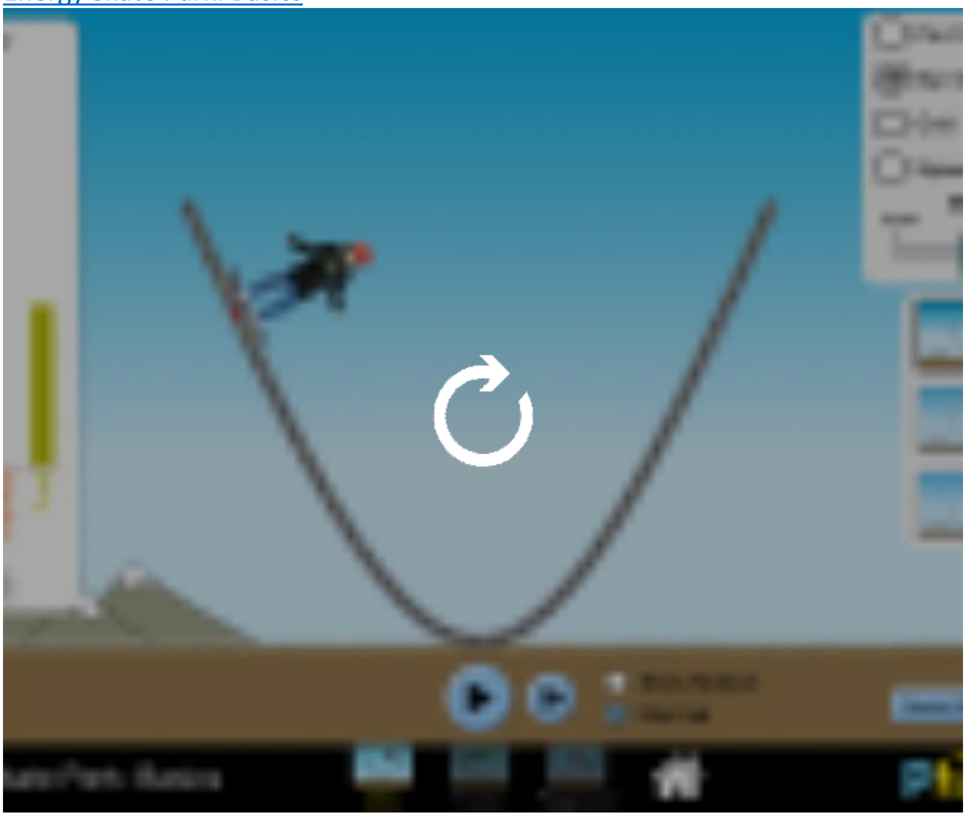
Videos and Simulations

Freitag, 4. April 2025 06:30

Videos:
Potential and kinetic energy:
https://app.jove.com/embed/player?id=12665&access=467dd12a10&t=1&s=1&fpv=1
Kinetic Energy:
https://app.jove.com/embed/player?id=12666&access=4d1e6cd988&t=1&s=1&fpv=1
Conservation of Energy:
https://app.jove.com/embed/player?id=14702&access=46e888c35b&t=1&s=1&fpv=1
Power:
https://app.jove.com/embed/player?id=12669&access=876e1894e9&t=1&s=1&fpv=1

The applets don't really work well inside OneNote, it is better to click on the links and open them in your browser.

Energy Skate Park: Basics



Trampoline (does not seem to work in school Wifi)
https://interactives.ck12.org/simulations/physics/trampoline/app/index.html?screen=sandbox

Worksheets

Mittwoch, 17. Juni 2026 06:19

Worksheet 1

Mittwoch, 17. Juni 2026 06:19

5. Work,
Energy, a...

5. Work, Energy, and Power: Worksheet 1

Ropes and Pulleys

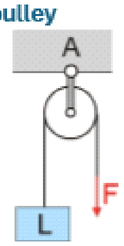
You certainly remember back from "Natur & Technik" that one can use clever combinations of ropes and pulleys to lift heavy loads. Let in all the following situations $m = 6\ \mathrm{kg}$, so the load is pulled down by the weight force (gravity) $L = m \cdot g = 60\ \mathrm{N}$.

Why such systems work is simple: the weight force L distributes evenly to all rope pieces going to L, which then is the required force F. **Remember that due to NEWTON's 1st law, pulling the load upwards with constant velocity requires the <u>same</u> force as just holding it (both are constant motions!), if friction in the system is negligible.**

In the following examples, find the force F necessary as well as the suspension forces F_A and F_B. Take the following steps:

1. $n =$ number of pieces of rope going to load L
2. force in rope (tension): $F = \frac{L}{n}$
3. $F_\mathrm{A} = F \times$ number of pieces of rope going to A (same with B)
4. cross check with Newton's 1st law: $\vec{F}_\mathrm{net} = \vec{0}$, hence the sum of all upwards forces must equate the sum of all downwards forces in the system.

fixed pulley

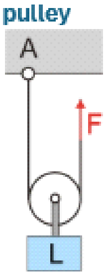


$n =$

$F =$

$F_\mathrm{A} =$

moving pulley

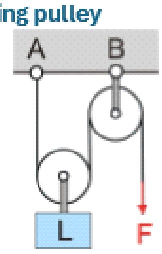


$n =$

$F =$

$F_\mathrm{A} =$

combination of fixed and moving pulley



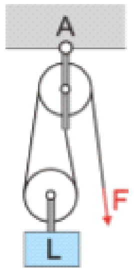
$n =$

$F =$

$F_\mathrm{A} =$

$F_\mathrm{B} =$

"gun tackle"

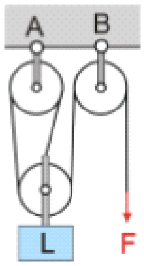


$n =$

$F =$

$F_A =$

"luff tackle"



$n =$

$F =$

$F_A =$

$F_B =$

The solutions for the forces got out of order (cross them as you go):

$$\frac{L}{3}, \frac{L}{2}, \frac{L}{2}, \frac{L}{2}, \frac{L}{2}, \frac{L}{2}, \frac{2}{3}L, \frac{2}{3}L, \frac{3}{2}L$$

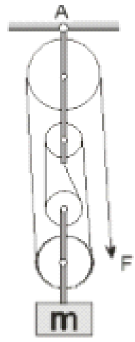
However, there is a trade-off: if the load rises by height h, *all* n pieces of rope going to L must be shortened by h, hence the rope length required to pull is

$$s = n \cdot h$$

In above examples, add the rope length that must be pulled to lift the load by $1\ \mathrm{m}$!

What we save in force we need to compensate in distance. This is the **Golden Rule of Mechanics**.

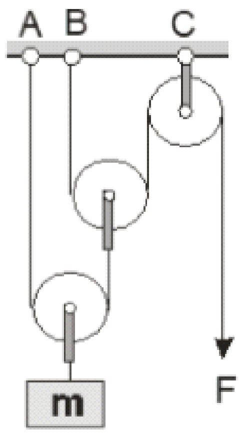
Additional Pulley Problems



"double tackle"[1]

a) What force F is necessary to hold the mass m?
b) What force acts on the suspension hook A in the ceiling?

$m = 13\ \mathrm{kg}$, small pulley $1\ \mathrm{kg}$, medium pulley $2\ \mathrm{kg}$, big pulley $3\ \mathrm{kg}$



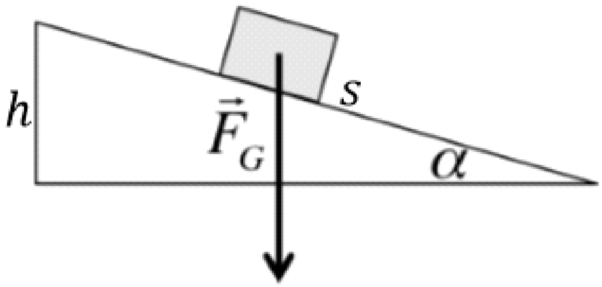
"Potenzflaschenzug"[2]

a) What force F is necessary to hold the mass m?
b) What force act at the points A, B, C in the ceiling?

$m = 12\ \mathrm{kg}$, pulleys $2\ \mathrm{kg}$

[1] a) $40\ \mathrm{N}$ b) $240\ \mathrm{N}$
[2] a) $45\ \mathrm{N}$ b) $70\ \mathrm{N}$, $45\ \mathrm{N}$, $110\ \mathrm{N}$

Inclined Plane Again



Another way to bring a heavy load to a point higher by h is using a ramp. Remember: to push something up a frictionless plane, we (at least) need to overcome the downhill force

$$F_{\mathrm{G}\parallel} = m \cdot g \cdot \sin\alpha \ .$$

Let us first have another look at the "beer barrel" problem on the last page of the last mock exam. Without ramp, it would require $1200\ \mathrm{N}$ of force to lift a barrel over a distance of $h = 1\ \mathrm{m}$ vertically.

a) Using the ramp, by what factor did the required force decrease? By what factor did the distance increase?

b) Using trigonometry, now find a general expression for the length of the ramp s involving h and α.

c) Demonstrate that the golden rule applies here as well for any $\alpha > 0$.

The golden rule does not apply anymore strictly if friction is involved, since then we additionally need to overcome the kinetic friction force. This gets only worse with smaller angles α since then

$$F_{\mathrm{kf}} = \mu_{\mathrm{k}} \cdot F_{\mathrm{N}} = \mu_{\mathrm{k}} \cdot m \cdot \cos\alpha$$

gets stronger and stronger!

Worksheet 2

Mittwoch, 17. Juni 2026 07:18

5. Work,
Energy, a...

1. Give a short answer:
   a) What is the definition of work in physics?
   b) The net force on an object is zero: what is the net work done on the object?
   c) What is the work done by the centripetal force on a rotating object?
   d) What type of quantity is one kilowatt hour?
   e) Give a simple example for one kilowatt hour.

2. Energy consumption of appliances:
   a) Name two quantities that determine the energy consumption of an electrical appliance.
   b) For each of the two above found quantities sort the following appliances qualitatively (big, medium, or small): kitchen fridge, hot plate, electrical boiler (for an apartment), and hairdryer.

3. You carry a heavy backpack of mass 15 kg.
   a) Calculate the gravitational force of the backpack.
   b) What work is needed at least (without friction) to lift the backpack 100 m in altitude? Both, formula, and numeric result are wanted.
   c) If the lifting is done in 10 min, what is the average power?

4. A retaining lake ("Stausee") has an average depth of 50 m, an average width of 200 m, and a length of 1 km.
   a) Calculate the mass of the retained water.
   b) Let's assume an electricity station is located 200 m below the lake and it can convert the stored energy without any losses into electricity. How many kilowatt hours can it produce by emptying the entire lake?

5. A cyclist has a mass of $m = 68\,\mathrm{kg}$, bicycle included. Assume the average power of the legs to be 90 W. What is the time needed to mount a difference in altitude of 450 m?

6. A energy bar (say a snickers or the like) has all nutrition facts listed on its wrapper. Energy is indicated as 1871 kJ. Under the assumption that one quarter of this energy can be used to climb a stairs, what is the height a 60 kg person could climb upstairs? Solve first the formula for h, then insert the numbers.

7. A crane lifts up a (metric) ton of concrete with a constant velocity of $v = 0.4\,\mathrm{m/s}$. Any frictional forces shall be not taken into account. Please insert the numbers just for the last part.
   a) Calculate the difference in height after a time interval Δt.
   b) Which work did the crane on the concrete during this time span?
   c) What is the power of the crane?
   d) What is the power of the crane measured in kW?

8. A stone of known mass is dropped from rest.
   a) What work did the gravitational force do on the stone, when the falling distance of the stone is Δh?
   b) Equalize this work with the kinetic energy that is given by $W_{kin} = 1/2 \cdot m \cdot v^2$ and calculate from this the speed of the stone as a function of the falling distance (no air friction).

9. A crate has a mass of 50 kg. It is slid on a horizontal floor. A kinetic friction coefficient of $\mu_k = 0.3$ is applicable.
   a) Calculate the friction force.
   b) Which work needs to be done in order to move the crate 8 m away?
   c) What power is needed when the velocity of the displacement is 0.2 m/s?

10. Consider two bodies with their given masses m_1 and m_2. Both bodies have the same kinetic energy.
    a) Calculate the ratio of their speeds $\frac{v_1}{v_2}$.
    b) Assume body 1 is a CO_2 molecule with 44 atomic mass units and body 2 is a N_2 molecule with 28 atomic mass units. Both molecules have the same kinetic energy. Calculate the speed of the CO_2 molecule when the speed of the N_2 molecule is 500 m/s.

11. A spring is stretched by $\Delta l = 10\,\text{cm}$. The amount of work done on the spring was 10 J.
    a) Draw a qualitative diagram force vs. elongation of the spring. Draw into this diagram the stretching work done.
    b) Which amount of work is needed to expand the spring another 10 cm? In the previous diagram draw in this amount of work and calculate its value.
    c) Calculate the spring constant D of the spring.

12. A boat of mass 200 kg travels with constant speed of 2 m/s. The outboard engine has a power of 3 kW.
    a) Calculate the amount of work the engine does in one second.
    b) Calculate the friction force of the water.
    c) Derive a general formula for the unknown force.

13. A car has a total mass (including all persons) of 1500 kg. It drives on a horizontal road with a speed of 60 km/h and needs to fully brake. The net braking distance (without reaction distance) is 36 m.
    a) What happens from an energy point of view?
    b) What is the average braking force?
    c) Under the same conditions: how long is the net braking distance for a car driving with double the speed (120 km/h)?

14. The string of a contrabass is plucked in its middle point. In very good approximation it holds true that a doubling of the plucking force leads to a doubling of the elongation (= displacement of the midpoint of the string from its position at rest).
    a) Tell a similar experiment known from physics class that can be compared to the situation of the string.
    b) Draw a qualitative diagram applied plucking force versus the elongation of the midpoint.
    c) How can the amount of work be visualized in this diagram?
    d) It is assumed that just before releasing the string a force of 3 N led to an elongation of 2.5 cm. Calculate the amount of work the finger did do on the string.

15. Two apples are hanging on an apple tree, located h_1 and h_2 above ground. Their masses are given as m_1 and m_2. When they fall from the tree they hit the ground with the speeds v_1 and v_2. Air friction shall be neglected.
    a) What is the ratio of their maximum kinetic energies?
    b) What is the ratio of their maximum speeds?

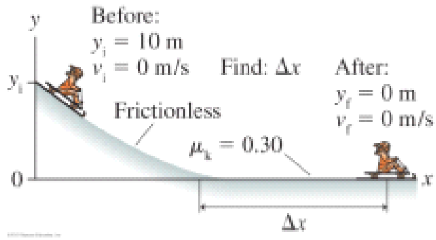
16. Within 10 min, a pump facility is able to lift an amount of $60\,\text{m}^3$ water from a well of 7 m depth. The electric engine consumes an electric power of 11.5 kW and has an efficiency of $\eta_{\text{motor}} = 0.85$, which means that 85 % of the electric power is transferred to the pump.

    a) Calculate the power transferred to the pump.

    b) Calculate the efficiency of the pump.

    c) What is the overall efficiency of the pumping facility

17. A sledder, starting from rest, slides down a 10 m-high hill. At the bottom of the hill is a long horizontal patch of rough snow. The hill is nearly frictionless, but the coefficient of friction between the sled and the rough snow at the bottom is $\mu_k = 0.3$.



    a) Which type of energies are involved during the down hill phase?

    b) Which energies are involved in the rough patch phase?

    c) Calculate the velocity of the sledder at the end of the down hill phase, first as formula and then in m/s.

    d) What is the friction force on the horizontal patch for a given mass m?

    e) How far will the sled slide along the rough patch?

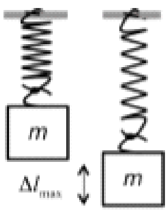
18. A wind-up car (this is a toy) has a mass of 20 g. A maximum speed of 2 m/s is achieved.

    a) What amount of energy can the spring of the car store at least?

    b) Which height can the car reach when it drives up a ramp? Assume friction-less conditions, establish a formal equation and solve for the height. Plugin the numbers.

    c) Assume, a second wind-up car is identical to the first one, except of the spring constant, which is doubled with respect to the first one. Under identical wind-up conditions: by which factor would the maximum speed change?

19. A piece of mass of 0.2 kg is attached to a relaxed spring. When the weight is released is stretches the spring to a maximum elongation Δl_{max}. At this maximum elongation, the mass will change its direction of the velocity and move up again.



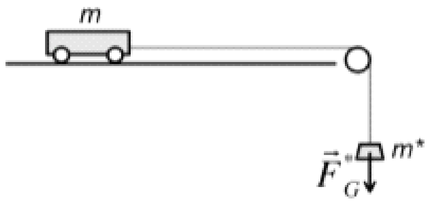
The spring constant is given as $D = 30\,\text{N/m}$.

    a) Define the reference level at the lowest position of the weight and calculate all energies at the point of time when the weight is released.

    b) The same for the lowest position.

    c) Calculate Δl_{max} by a formula and by numbers.

20. A cart can move on a horizontal plane nearly without friction. The cart is attached to a cord which is connected via a pulley to a weight. The pulley's mass is negligible with respect to the weight and the cart. The cart is released from rest and starts to move. After a certain time it is observed that the cart is displaced by Δx .



a) Establish all energies at the initial state and all energies at the final state.

b) Calculate the velocity of the cart as a function of Δx .

21. Proper design of automobile braking systems must account for heat buildup under heavy braking. Assume a 1500 kg car that descends a 20° hill. The car begins braking when its speed is 90 km/h and slows to a speed of 30 km/h in a distance of 0.3 km measured along the road.

a) Formulate in words what determines the dissipated energy of the brakes.

b) Calculate the dissipated energy of the brakes.

c) For how long would a hot plate of 2 kW need to be on in order to dissipate the same thermal energy?

1. a) Work is the scalar product of force and displacement $\Delta W = \vec{F} \cdot \Delta \vec{s}$.
   b) The net work is zero as well.
   c) Work is zero because the centripetal force acts perpendicular to the displacement.
   d) Energy or work.
   e) Cooking for half an hour with a 2 kW hot plate consumes the energy of 1 kW h.

2. a) Time of usage and power: $\Delta W = \Delta t \cdot P$.
   b) time of usage: fridge big, boiler medium, hot plate short, hairdryer very short.
   power: boiler very big, hot plate big, hairdryer medium, fridge small.

3. a) $F_G = m \cdot g \approx 150\,\mathrm{N}$
   b) $\Delta W = m \cdot g \cdot \Delta h = 150\,\mathrm{N} \cdot 100\,\mathrm{m} = 15\,000\,\mathrm{J} = 15\,\mathrm{kJ}$
   c) $P = {}^{\Delta W}/_{\Delta t} = {}^{15\,000\,\mathrm{J}}/_{600\,\mathrm{s}} = 25\,\mathrm{W}$.

4. a) Volume: $V = l \cdot w \cdot d = 1 \times 10^7\,\mathrm{m}^3$
   Mass: $m = V \cdot \varrho = 1 \times 10^7\,\mathrm{m}^3 \cdot 1000\,\mathrm{kg/m}^3 = 1 \times 10^{10}\,\mathrm{kg}$
   b) $W = F_G \cdot \Delta h = m \cdot g \cdot \Delta h = 1 \times 10^{11}\,\mathrm{N} \cdot 200\,\mathrm{m} = 2 \times 10^{13}\,\mathrm{J} = 5.55 \times 10^6\,\mathrm{kW\,h} = 5.55\,\mathrm{GW\,h}$.
   The nuclear power plant Gösgen would need around 5.55 h to produce the same amount of energy.

5.

$$P = \frac{\Delta W}{\Delta t} \Rightarrow \Delta t = \frac{\Delta W}{P} = \frac{m \cdot g \cdot \Delta h}{P} = \frac{68\,\mathrm{kg} \cdot 10\,\mathrm{m/s}^2 \cdot 450\,\mathrm{m}}{90\,\mathrm{W}} = 3400\,\mathrm{s}$$

6.

$$\frac{W}{4} = m \cdot g \cdot \Delta h$$
$$\Delta h = \frac{W}{4 \cdot m \cdot g} = \frac{1871\,\mathrm{kJ}}{4 \cdot 60\,\mathrm{kg} \cdot 9.81\,\mathrm{m/s}^2} = 794.7\,\mathrm{m}$$

7. a) $\Delta h = v \cdot \Delta t$
   b) $\Delta W = F_G \cdot \Delta h = m \cdot g \cdot \Delta h = m \cdot g \cdot v \cdot \Delta t$.
   c) $P = {}^{\Delta W}/_{\Delta t} = m \cdot g \cdot v$, Δt cancels out.
   d) $P = 1000\,\mathrm{kg} \cdot 9.81\,\mathrm{m/s}^2 \cdot 0.4\,\mathrm{m/s} \approx 4000\,\mathrm{W} = 4\,\mathrm{kW}$.

8. a) $\Delta W = F_G \cdot \Delta h = m \cdot g \cdot \Delta h$
   b) $m \cdot g \cdot \Delta h = {}^1/_2 \cdot m \cdot v^2 \;\Rightarrow\; g \cdot \Delta h = {}^1/_2 v^2 \;\Rightarrow\; v = \sqrt{2 \cdot g \cdot \Delta h}$ independent of mass.

9. a) $F_f = \mu_k \cdot F_N = \mu_k \cdot F_g \approx 150\,\mathrm{N}$
   b) $\Delta W = F_f \cdot \Delta s = 150\,\mathrm{N} \cdot 8\,\mathrm{m} = 1200\,\mathrm{J}$.
   c) $P = \frac{\Delta W}{\Delta t} = F_f \cdot \frac{\Delta s}{\Delta t} = F_f \cdot v = 150\,\mathrm{N} \cdot 0.2\,\mathrm{m/s} = 30\,\mathrm{W}$

10.

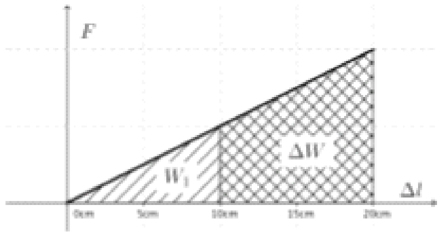
a)

$$\frac{1}{2} \cdot m_1 \cdot v_1^2 = \frac{1}{2} \cdot m_2 \cdot v_2^2$$
$$\frac{v_1^2}{v_2^2} = \frac{m_2}{m_1}$$
$$\frac{v_1}{v_2} = \sqrt{\frac{m_2}{m_1}}$$

b)

$$v_1 = \sqrt{\frac{m_2}{m_1}} \cdot v_2 = \sqrt{\frac{28\,\mathrm{u}}{44\,\mathrm{u}}} \cdot 500\,\mathrm{m/s} = 398\,\mathrm{m/s}$$

11. a)



b) The elongation is doubled, therefore the energy is multiplied by four. $\rightarrow +30\,\text{J}$ are needed.

c)

$$\begin{aligned} W_{\text{spring}} &= \frac{1}{2} \cdot D \cdot \Delta l^2 && \text{solve for } D \\ D &= \frac{2 \cdot W_{\text{spring}}}{\Delta l^2} = \frac{2 \cdot 10\,\text{J}}{0.1\,\text{m} \cdot 0.1\,\text{m}} = 2000\,\text{N/m} \end{aligned}$$

12. a)

$$\Delta W = P \cdot \Delta t = 3\,\text{kW} \cdot 1\,\text{s} = 3\,\text{kJ} = 3000\,\text{J}.$$

b) Every second leads to a displacement of 2 m.

$$\Delta W = F \cdot \Delta s \;\Rightarrow\; F = \frac{\Delta W}{\Delta s} = \frac{3000\,\text{J}}{2\,\text{m}} = 1500\,\text{N}.$$

c)

$$F = \frac{\Delta W}{\Delta s} = \frac{\Delta W}{v \cdot \Delta t} = \frac{P}{v}$$

13. a) Kinetic energy is converted to friction heat.

b)

$$\begin{aligned} W_{\text{f}} &= F_{\text{f}} \cdot \Delta s && \text{friction work} \\ W_{\text{kin}} &= \frac{1}{2} \cdot m \cdot v^2 && \text{kinetic energy} \\ \Rightarrow F_{\text{f}} &= \frac{W_{\text{f}}}{\Delta s} = \frac{W_{\text{kin}}}{\Delta s} = \frac{m \cdot v^2}{2 \cdot \Delta s} = \frac{1500\,\text{kg} \cdot (16.7\,\text{m/s})^2}{2 \cdot 36\,\text{m}} = 5787\,\text{N} \end{aligned}$$

c) Because the kinetic energy is proportional to the square of the speed, the kinetic energy is multiplied by four. So is the braking distance $4 \cdot 36\,\text{m} = 144\,\text{m}$.

14. a) Expanding a spring.

b) The force is directly proportional to the elongation $\Rightarrow$ straight line through the origin.

c) Work visualizes as area under the curve.

d) The average force is half of the maximum force.

$$\Delta W = F \cdot \Delta l = \frac{F_{\text{max}}}{2} \cdot \Delta l = \frac{3\,\text{N}}{2} \cdot 0.025\,\text{m} = 0.0375\,\text{J}$$

15. a) Potential energy is converted to kinetic energy. $W_{\text{kin}} = W_{\text{pot}}$

$$\Rightarrow \frac{W_{\text{kin},1}}{W_{\text{kin},2}} = \frac{W_{\text{pot},1}}{W_{\text{pot},2}} = \frac{m_1 \cdot g \cdot h_1}{m_2 \cdot g \cdot h_2} = \frac{m_1 \cdot h_1}{m_2 \cdot h_2}$$

b)

$$\begin{aligned} m \cdot g \cdot h &= \frac{1}{2} \cdot m \cdot v_{\text{max}}^2 && \text{mass cancels out} \\ \Rightarrow v_{\text{max}} &= \sqrt{2 \cdot g \cdot h} \\ \Rightarrow \frac{v_{\text{max},1}}{v_{\text{max},2}} &= \frac{\sqrt{2 \cdot g \cdot h_1}}{\sqrt{2 \cdot g \cdot h_2}} = \sqrt{\frac{h_1}{h_2}} \end{aligned}$$

16. a)

$$P_{\text{pump}} = \eta_{\text{motor}} \cdot P_{\text{elektrisch}} = 0.85 \cdot 11.5\,\text{kW} = 9.775\,\text{kW}$$

b)

$$\begin{aligned}\eta_{\text{pump}} &= \frac{P_{\text{Hub}}}{P_{\text{pump}}} = \frac{m \cdot g \cdot \Delta h}{P_{\text{pump}} \cdot \Delta t} \\ &= \frac{60 \times 10^3\,\text{kg} \cdot 10\,\text{m/s}^2 \cdot 7\,\text{m}}{9.775\,\text{kW} \cdot 600\,\text{s}} = 0.716\end{aligned}$$

c)

$$\begin{aligned}P_{\text{lifting}} &= \eta_{\text{pump}} \cdot P_{\text{motor}} = \eta_{\text{pump}} \cdot \eta_{\text{motor}} \cdot P_{\text{elektrisch}} = \eta_{\text{overall}} \cdot P_{\text{elektrisch}} \\ &\Rightarrow \eta_{\text{overall}} = \eta_{\text{pump}} \cdot \eta_{\text{motor}} = 0.716 \cdot 0.85 = 0.61 = 61\,\%\end{aligned}$$

17. a) Potential energy is converted into kinetic energy.

b) Kinetic energy is converted to friction heat.

c) Reference level is taken at the horizontal patch. Initial state (i): at rest, final state (f): at end of hill, moving with v_{f}

$$W_{\text{pot,i}} = m \cdot g \cdot h \qquad W_{\text{kin,i}} = 0$$

$$W_{\text{pot,f}} = 0 \qquad W_{\text{kin,f}} = \frac{1}{2} \cdot m \cdot v^2$$

$$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2 \Rightarrow v_{\text{f}} = \sqrt{2 \cdot g \cdot h} \approx 14.1\,\text{m/s}$$

d)

$$F_{\text{f}} = \mu_{\text{k}} \cdot F_{\text{N}} = \mu_{\text{k}} \cdot g \cdot m$$

e) Reference level is taken at the horizontal patch. Initial state (i): at rest, final state (f): on the rough part at rest.

$$W_{\text{pot,i}} = m \cdot g \cdot h \qquad W_{\text{kin,i}} = 0$$

$$W_{\text{pot,f}} = 0 \qquad W_{\text{kin,f}} = 0$$

$$W_{\text{friction}} = F_{\text{f}} \cdot \Delta x = \mu_{\text{k}} \cdot g \cdot m \cdot \Delta x$$

$$W_{\text{pot,i}} = W_{\text{friction}} \quad \Rightarrow \Delta x = \frac{h}{\mu_{\text{f}}}$$

18. a)

$$W_{\text{kin}} = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2} 0.020\,\text{kg} \cdot (2\,\text{m/s})^2 = 0.04\,\text{J}$$

b)

$$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2 \quad \Rightarrow \quad h = \frac{v^2}{2g} \approx 0.2\,\text{m}$$

c)

$$W_{\text{spring}} = \frac{1}{2} \cdot D \cdot (\Delta l)^2$$

Spring energy is doubled and so is the kinetic energy. Because the mass stays the same, the maximum speed increases by a factor of $\sqrt{2}$.

19. a)

$$W_{\text{kin,i}} = W_{\text{spring,i}} = 0 \ , W_{\text{pot,i}} = m \cdot g \cdot \Delta l_{\text{max}}$$

b)

$$W_{\text{kin,f}} = W_{\text{pot,f}} = 0 \ , W_{\text{spring,f}} = \frac{1}{2} \cdot D \cdot \Delta l_{\text{max}}^2$$

c)

$$\begin{aligned}W_{\text{pot,i}} &= W_{\text{spring,f}} \\ m \cdot g \cdot \Delta l_{\text{max}} &= \frac{1}{2} \cdot D \cdot \Delta l_{\text{max}}^2 \\ \Delta l_{\text{max}} &= \frac{2 \cdot g \cdot m}{D} \approx \frac{4\,\text{N}}{30\,\text{N/m}} = 0.133\,\text{m}\end{aligned}$$

20. The reference level is taken at the final location of the weight. When the cart is displaced by Δx, the weight is vertically displaced by the same distance. Both masses have always the same velocity.

a)

$$W_{\text{pot,i}} = m^* \cdot g \cdot \Delta x \qquad W_{\text{kin,i}} = 0$$
$$W_{\text{pot,f}} = 0 \qquad W_{\text{kin,f}} = \frac{1}{2} \cdot (m + m^*) \cdot v_f^2$$

b)

$$m^* \cdot g \cdot \Delta x = \frac{1}{2} \cdot (m + m^*) \cdot v_f^2$$
$$v_f = \sqrt{\frac{m^*}{m + m^*} \cdot 2 \cdot g \cdot \Delta x}$$

21. a) The speed is reduced, hence the kinetic energy is decreased. In addition, the potential energy of the car is reduced. The dissipated energy of the brake is the sum of these two energy differences.

b)

$$\Delta W_{\text{pot}} = m \cdot g \cdot \Delta h = m \cdot g \cdot s \cdot \sin\alpha \approx 1500\,\text{kg} \cdot 10\,\text{m/s}^2 \cdot 300\,\text{m} \cdot \sin 20^\circ = 1.54\,\text{MJ}$$
$$\Delta W_{\text{kin}} = \frac{1}{2} \cdot m \cdot (v_1^2 - v_2^2) = \frac{1}{2} \cdot 1500\,\text{kg} \cdot ((\frac{90}{3.6}\text{m/s})^2 - (\frac{30}{3.6}\text{m/s})^2) = 0.42\,\text{MJ}$$

The total dissipated thermal energy is 1.96 MJ.

c)

$$P = \frac{\Delta W}{\Delta t} \quad \Rightarrow \quad t = \frac{\Delta W}{P} = \frac{1.96 \times 10^6\,\text{J}}{2 \times 10^3\,\text{W}} = 980\,\text{s} = 16.3\,\text{min} \quad (!).$$

Worksheet 3

Mittwoch, 17. Juni 2026 06:21

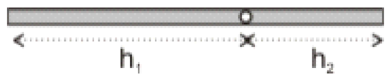
5. Work, Energy, a...

5. Work, Energy, and Power: Worksheet 3

Law of the Lever (*Hebelgesetz*)

Simple machines such as ropes and pulleys (worksheet 1) or levers take in an input work (i.e. a force acting over some distance) and again output mechanical work on the other side. If we neglect frictional losses (which would be some input work converted to mostly useless heat energy) and if the machine does not store any energy (e.g. in a spring), **Conservation of Energy** commands that input and output work must be exactly equal, which in this special case then is also called **Golden Rule of Mechanics**. For levers, we can use similar triangles to replace the (variable) distances by the (fixed) lengths of the lever arms h_1 and h_2:

$$\begin{aligned} \text{input work} &= \text{output work} \\ \Delta W_1 &= \Delta W_2 \\ F_1 \cdot s_1 &= F_2 \cdot s_2 \end{aligned}$$



Hence we end up with the **Law of the Lever**:

two-sided lever: pivot (*Drehpunkt*) is <u>between</u> the lever arms[1]

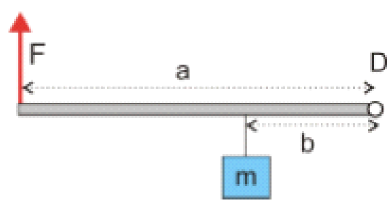
What mass m_2 needs to be attached such that the board does not tilt? In this and all the following problems, the mass of the lever (boards) itself is to be neglected.

$m_1 = 80\ \mathrm{kg}$, length of the board $L = 6\ \mathrm{m}$, $a = 2\ \mathrm{m}$

one-sided lever: both lever arms extend to the <u>same</u> side of the pivot[2]

What force F needs to be applied such that the board does not tilt?

Important: The length of lever arms is always measured between the point where the force attacks and the **pivot**! $m = 15\ \mathrm{kg}$, $a = 50\ \mathrm{cm}$, $b = 20\ \mathrm{cm}$

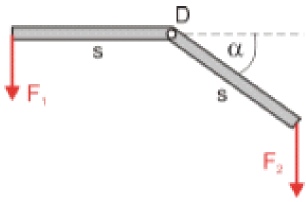


[1] 40 kg
[2] 60 N

angled lever[3]

How strong does F_2 need to be such that the board does not tilt?

$F_1 = 200\ \mathrm{N}\,,\, s = 4\ \mathrm{m}\,,\, \alpha = 45°$



The straight line that a force vector lies on is called **line of action**. The relevant lever arm length is always the **shortest** distance between pivot and line of action (and thereby must meet it perpendicularly)!

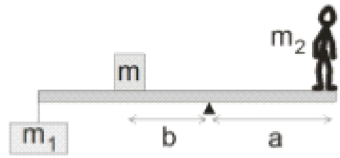
more than two forces[4]

What mass m is needed on the board for balance?

length of the board $L = 3\ \mathrm{m}\,,\, a = 1\ \mathrm{m}\,,\, b = 50\ \mathrm{cm}\,,\, m_1 = 25\ \mathrm{kg}\,,\, m_2 = 60\ \mathrm{kg}$

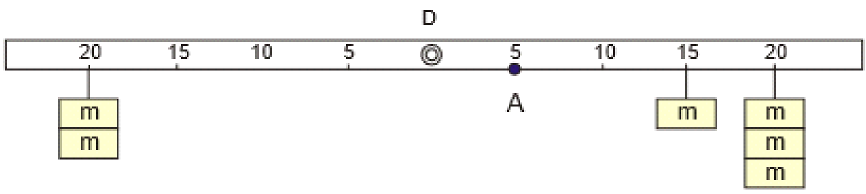
In a situation like this, the Law of the Lever extends to:

$$\underbrace{F_1 \cdot h_1 + F_2 \cdot h_2 + F_3 \cdot h_3 + \cdots}_{\text{left side}} = \underbrace{F_4 \cdot h_4 + F_5 \cdot h_5 + F_6 \cdot h_6 + \cdots}_{\text{right side}}$$



Exam Problem 24/25: Measuring Beam[5]

Mass pieces of $m = 50$ g each are suspended from a 50 cm long measuring beam with pivot D in the middle as shown in the sketch.



a) How much force F is required at point A for balance?

b) Can this balance be achieved by hanging some mass piece from A?
Hint: You should have obtained a negative value of for F in a). What could that mean?

[3] $283\ \mathrm{N}$

[4] $20\ \mathrm{kg}$

[5] a) $F = -3.5\ \mathrm{N}$ b) The minus sign indicates that this force must be upwards and hence cannot be caused by a mass piece since gravity always acts downwards. Requires for example a string from the ceiling.

Worksheet 4

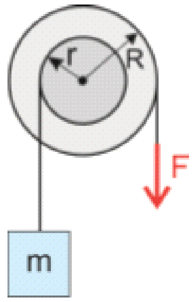
Mittwoch, 17. Juni 2026 07:24

5. Work, Energy, a...

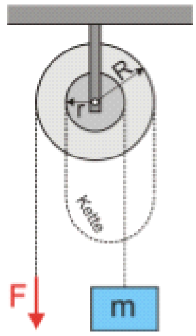
5. Work, Energy, and Power: Worksheet 4

Wheel and Axle, Screws

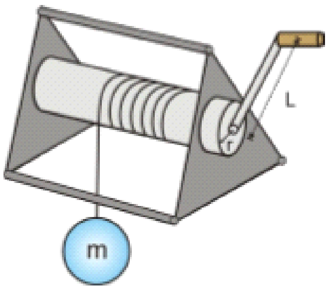
- Wheel and axle: m = 1.2 kg, r = 30 cm, R = 40 cm
  How much force F is needed to lift the mass m?



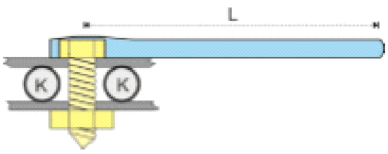
- Differential pulley: m = 12 kg, r = 15 cm, R = 20 cm
  How much force F is needed to lift the mass m?



- Wheel and axle: L = 30 cm, m = 36 kg, r = 5 cm
  How much force F is needed to turn the crank to lift the lead ball m?



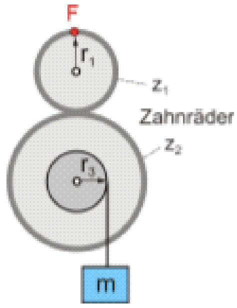
- Screw: pitch h = 1mm, L = 20 cm, F_1 = 10 N
  With what force F_2 are the balls K crushed if the end of the wrench is turned with F_1?



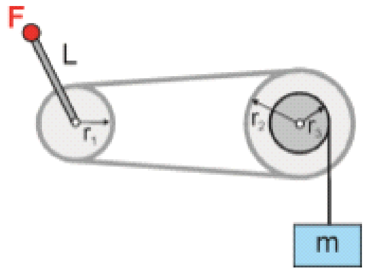
not in order: 12.6 kN, 9 N, 90 N, 60 N

Gears, Chains

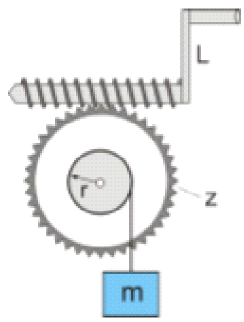
- Gears:
  $m = 1.2$ kg, $r_1 = 10$ cm, $z_1 = 40$, $z_2 = 80$ $r_3 = 5$ cm
  How much force F is needed to lift the mass m?
  Also draw the direction of F!



- Chain drive:
  $m = 1.2$ kg, $L = 40$ cm, $r_1 = 20$ cm, $r_2 = 30$cm, $r_3 = 10$ cm
  How much force F is needed to lift the mass m?
  Also draw the direction of F!



- Worm gear:
  $L = 30$ cm, $m = 48$ kg, $r = 20$ cm, $z = 40$
  How much force F is needed to lift the mass m, neglecting friction?
  Also draw the direction of F!

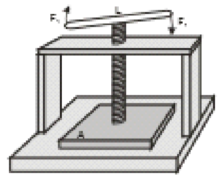


not in order: 2 N, 3 N, 8 N

Additional Problems

PRINTING PRESS

The crank L = 40 cm is turned with F_1 = 25 N at **both** ends. The pitch of the screw is h = 5 mm.
With what force F_2 is the plate A pressed onto the base?

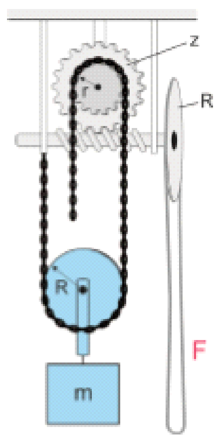


12.6 kN

SPECIAL PULLEY

mass m including pulley R = 120 kg, z = 20, r = 10 cm, R = 20 cm

How much force F is needed to lift the mass m?
Also draw the direction of F!



15 N

Objectives

Sonntag, 21. Juni 2026 10:53

Objectives
Work, En...

Scope

5 "Work, Energy, and Power", worksheets 1–4.

worksheet 2: without problems 2, 10, 15

worksheet 4: "Wheel and Axle, Screws" only (p. 1 and "Printing Press" problem on p. 3 only)

Learning Objectives: You know, understand, or are able to

General

1. the abbreviations and meaning of nano, micro, milli, centi, deci, kilo, mega and giga as unit prefixes.
2. scientific notation and how it is applied, as well as how to use it with your calculator.
   a) convert between prefixes and scientific notation.
3. do symbolic calculations.
4. do numerical calculations including the units at each step.

Main Topics

5. from forces: gravitational force (weight) $F_\mathrm{G} = m \cdot g$, kinetic friction force $|\vec{F}_\mathrm{f,k}| = \mu_\mathrm{k} \cdot |\vec{F}_\mathrm{N}|$, Hooke's law $F_\mathrm{spring} = k \cdot s$.
6. the definition of work as scalar product of force and displacement:
   $$\Delta W = \vec{F} \cdot \Delta\vec{s} = |\vec{F}| \cdot |\Delta\vec{s}| \cdot \cos\varphi$$
7. (that energy is stored work.)
8. kinetic friction work $\Delta W_\mathrm{kf} = F_\mathrm{f,k} \cdot \Delta s$, why static friction does no work.
9. potential energy $E_\mathrm{pot} = m \cdot g \cdot h$, kinetic energy $E_\mathrm{kin} = \frac{1}{2} \cdot m \cdot v^2$, spring energy $E_\mathrm{spring} = \frac{1}{2} \cdot k \cdot s^2$.
10. what conservation of energy is and how to apply it.
11. the definition of power $P = \frac{\Delta W}{\Delta t}$ and that this also implies $P = F \cdot v$.
12. units: especially (you still need to know the previous ones of course): force (N), work and energy (J, Ws, kWh and how they interconvert) and power (W, kW).
13. simple machines:
    a) why simple machines (without friction) are a prime example of the law of energy conservation (Golden Rule!) and that there is always a trade-off between force and distance.
    b) calculate forces in a rope and pulley system.
    c) apply the law of the lever.
    d) solve exercises including levers, pulleys, wheel and axle and screws.

Aids: You need pens of different colours, a ruler, a geo-triangle and your pocket calculator. You do not need your own writing paper. Provided on the exam sheet will be the most important equations, but without context or explanations of any kind.

Also check out the new mock exam on OneNote and the exam coach on fobizz!

Mock Exam 4

Sonntag, 21. Juni 2026 10:44

Mock Exam 4

Mock Exam: Work, Energy, and Power

Name: ____ Points: ____________ /27.5 Grade: ________

Time: 90 min
Aids: calculator

- **Write directly on the exam sheets. Use the back of the pages if necessary.**
- **Write down all the steps. You will not get any points if you only write down the result!**
- **Calculate symbolically as long as possible, only plug in numbers at the very last step.**
- **Round to three significant digits. Use prefixes or scientific notation where appropriate.**
- **<u>Use units all along your calculations, not just in the final result!</u>**
  You will not get full points otherwise.

Good Luck!

1. Child on a Sledge 2\

A child on a sledge is pulled across a snow-covered horizontal road. The angle of the rope is $40°$ with respect to the road. The mother pulls with a force of $15\ \mathrm{N}$.

a) Calculate the mother's work done for a displacement of $10\ \mathrm{m}$. /1

b) Determine the friction force. /1

2. The Worth of Human Work 4\

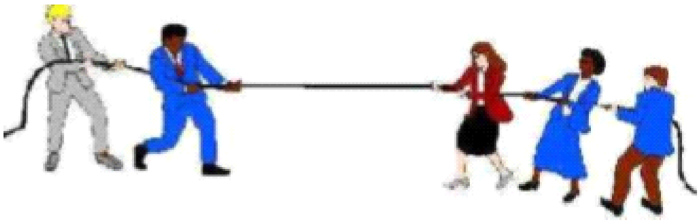
Over a normal work shift, manual workers can sustain an output of around $75\ \mathrm{W}$ of power. A worker continuously lifts wooden boards onto the load beds of trucks for one hour.

a) Calculate the work done in J and kWh. /1

b) What is the equivalent monetary worth of the work done, based on the cost of electric energy of currently about $30\ \mathrm{Rp./kWh}$? /1

c) A board weighs about $10\ \mathrm{kg}$ and the load beds are at an average height of $1.8\ \mathrm{m}$. How many boards can the worker load during this time? /2

3. Rope Pulling 4\



In contrast to hemp ropes, nylon ropes are highly elastic and store the work performed during stretching like a spring. If a rope breaks, this energy is suddenly released and accelerates the rope ends at the break points to high speeds. This has led to serious accidents in rope pulling competitions in the past, read the article below (not required to solve the problem).

a) What is the spring constant of a rope that is stretched by $5\ \mathrm{m}$ with a maximum force of $50\ \mathrm{kN}$? /1

b) What energy is stored in the rope? /1

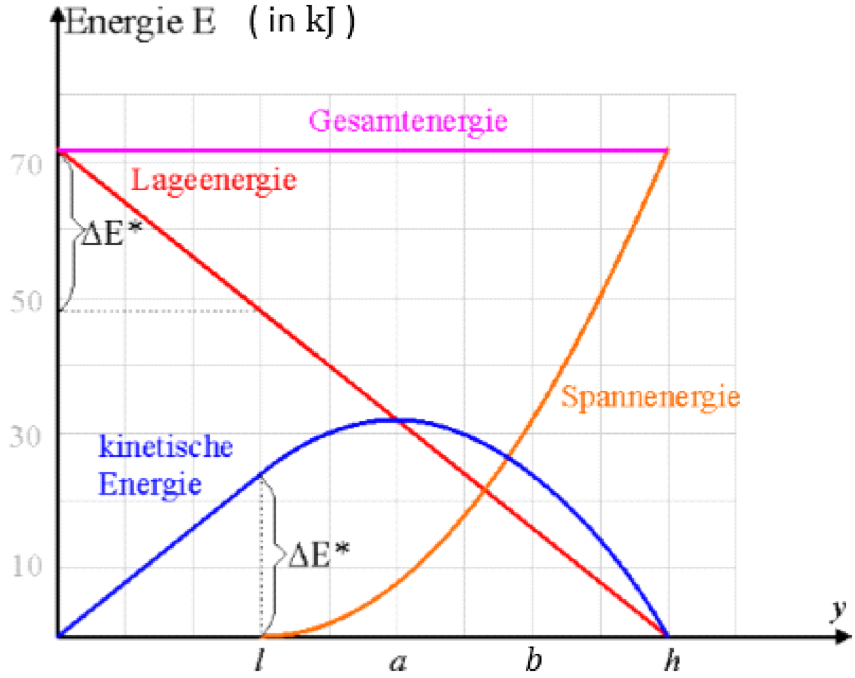
c) Assume this energy distributes evenly to both loose ends of the rope when it breaks, which are both assumed to be $1\ \mathrm{kg}$. Calculate the speeds they can reach and compare with the speed of sound in air ($330\ \frac{\mathrm{m}}{\mathrm{s}}$). /2

Zwei Kinder nach Tauziehen umgekommen

Nach einem Unfall bei einem Seilziehen in Westernohe (Deutschland) ist in der Nacht zum Dienstag ein zweites Kind seinen Verletzungen erlegen. Wie die Polizei mitteilte, stammte der zehnjährige Junge wie schon das erste Opfer aus Koblenz. Bei dem Tauziehen von 650 Jugendlichen in einem Pfadfinderlager war am Pfingstsonntag das daumendicke Nylonseil gerissen und hatte einen Buben auf der Stelle erschlagen. 24 Pfadfinder wurden verletzt, fünf von ihnen schwer. (ap, 1995)

4. Bungee Jumping 4.5\

The diagram displays the different forms of energy that occur during a bungee jump. The horizontal axis shows the depth of the jumper ($y = 0$: jump platform; $y = h$: lowest point).



a) During the jump, what is happening at the depths indicated by $y = l$ and $y = a$ or why are those depths special? /1

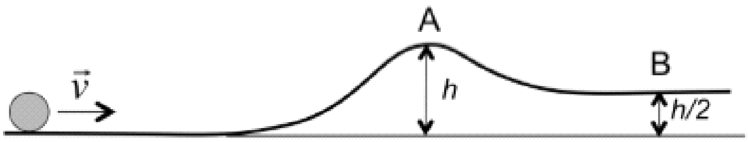
For the following two subexercises, read off the required data from the diagram as accurately as possible.

b) How does the curve of total energy (*Gesamtenergie*) relate to the three other curves? Confirm your assumption numerically for height b. /1.5

c) What is the maximum speed reached by a jumper weighing 60 kg? /2

5. Mini-Golf 4\

The picture shows a mini-golf ball with given mass m that moves with the initial speed $|\vec{v}|$ towards an obstacle. For simplicity, consider the motion to be *frictionless*.



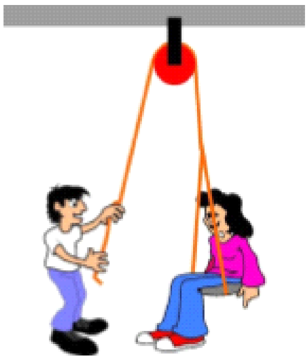
a) When the ball arrives at point A, its speed $|\vec{v}_A|$ is almost zero. Derive a formula for the initial speed $|\vec{v}|$ depending on h. /1

b) Plug in the numbers $m = 15$ g, $h = 30$ cm to obtain the numeric result for $|\vec{v}|$. /1

c) Which percentage of the initial speed does the ball have when it arrives at point B? /2

6. Dr Hansjakobli und ds Babettli 4\

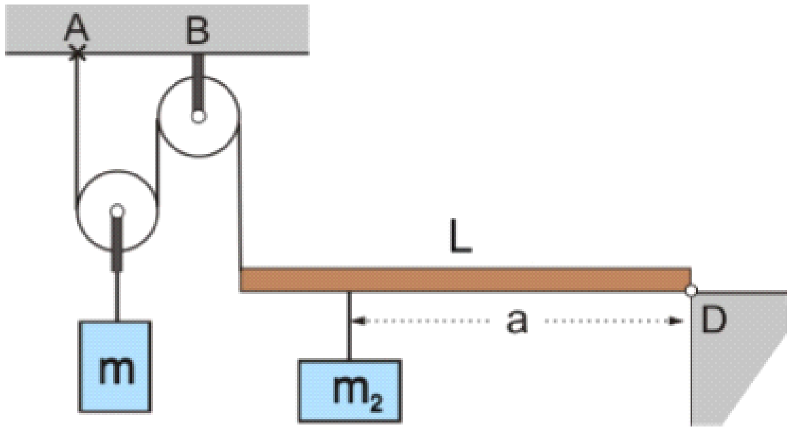
Babettli (40 kg) sits on a board to which a rope is attached, that runs over a pulley at the ceiling.



a) What force does Hansjakobli have to exert to pull Babettli up? What is the minimum force that the suspension on the ceiling must be able to hold? /2

b) Now Hansjakobli hands Babettli the end of the rope. Does this change the situation? Can she pull herself up and if so, what force does she need to exert? What is the minimum force that the suspension on the ceiling must be able to hold now? /2

7. Pulley and Lever Combination 5\



Given: $L = 1\ \mathrm{m}$, $a = 80\ \mathrm{cm}$, $m_2 = 300\ \mathrm{g}$, mass of the pulleys 100 g each.
The lever can be considered massless.

a) Calculate the force with which the lever pulls on the rope. /1
b) Calculate the mass m needed for equilibrium. /2
c) Calculate the forces acting at the points A and B at the ceiling. /2

Solutions

Sonntag, 21. Juni 2026 10:45

Mock Exam 4 Sol

Mock Exam: Work, Energy, and Power

Name: ______________________________ Points: __________ /27.5 Grade:______

Time: 90 min
Aids: calculator

- **Write directly on the exam sheets. Use the back of the pages if necessary.**
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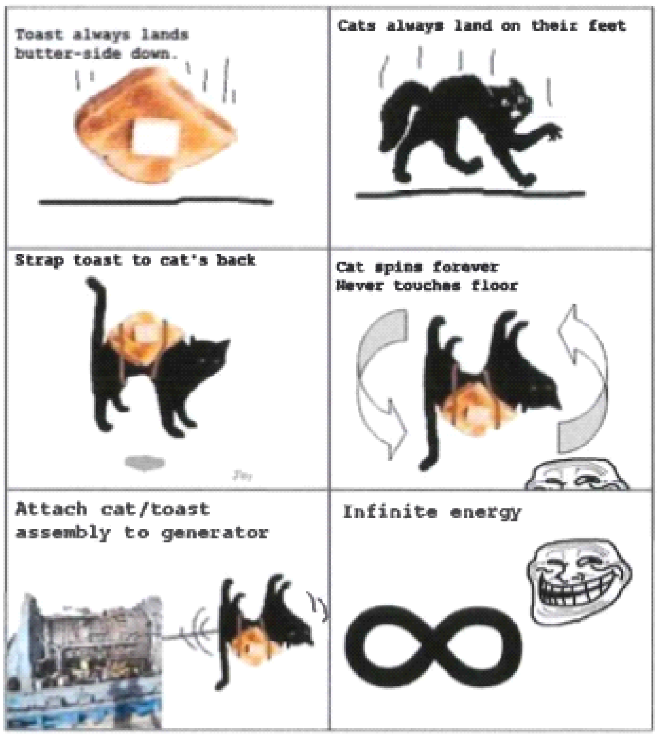
Good Luck!



page 1 of 6

1. Child on a Sledge

2\

A child on a sledge is pulled across a snow-covered horizontal road. The angle of the rope is 40° with respect to the road. The mother pulls with a force of 15 N.

a) Calculate the mother's work done for a displacement of 10 m. /1

b) Determine the friction force. /1

a) $\Delta W = F \cdot \Delta s \cdot \cos\varphi = 15\,N \cdot 10\,m \cdot \cos(40°) \approx 114.91\,J$

b) $F_{kf} = \frac{\Delta W}{\Delta s} = \frac{F \cdot \cancel{\Delta s} \cdot \cos\varphi}{\cancel{\Delta s}} = F \cdot \cos\varphi = 15\,N \cdot \cos(40°) \approx 11.5\,N$

2. The Worth of Human Work

4\

Over a normal work shift, manual workers can sustain an output of around $\overset{P}{75}$ W of power. A worker continuously lifts wooden boards onto the load beds of trucks for $\underset{\Delta t}{\text{one hour}}$.

a) Calculate the work done in J and kWh. /1

b) What is the equivalent monetary worth of the work done, based on the cost of electric energy of currently about 30 Rp./kWh ? /1

c) A board weighs about 10 kg and the load beds are at an average height of $\underset{\Delta h}{1.8}$ m. How many boards can the worker $\overset{m}{\text{load}}$ during this time? /2

a) $\Delta W = P \cdot \Delta t = 75\,W \cdot 3600\,s = 270\,000\,J = 2.7 \times 10^5\,J = 270\,kJ$

$= 0.075\,kW \cdot 1h = 0.075\,kWh = 7.5 \times 10^{-2}\,kWh$

b) $7.5 \times 10^{-2}\,kWh \cdot 30\,\frac{Rp.}{kWh} = 2.25\,Rp.$

c) $\Delta W = n \cdot m \cdot g \cdot \Delta h = n \cdot 180\,J$

$n = \frac{\Delta W}{m \cdot g \cdot \Delta h} = 1500$

3. Rope Pulling 4\

In contrast to hemp ropes, nylon ropes are highly elastic and store the work performed during stretching like a spring. If a rope breaks, this energy is suddenly released and accelerates the rope ends at the break points to high speeds. This has led to serious accidents in rope pulling competitions in the past, read the article below (not required to solve the problem).

a) What is the spring constant of a rope that is stretched by $\overset{s}{5}$ m with a maximum force of $\overset{F}{\text{50 kN}}$? /1

b) What energy is stored in the rope? /1

c) Assume this energy distributes evenly to both loose ends of the rope when it breaks, which are both assumed to be 1 $\overset{m}{\text{kg}}$. Calculate the speeds they can reach and compare with the speed of sound in air (330 $\frac{m}{s}$). /2

a) $F = k \cdot s \Rightarrow k = \frac{F}{s} = 10 \frac{kN}{m} = 10000 \frac{N}{m}$

b) $E_{spring} = \frac{1}{2} k \cdot s^2 = 125 \text{ kJ} = 1.25 \times 10^5 \text{ J}$

c) $\frac{1}{2} E_{spring} = \frac{1}{2} m \cdot v^2 \Rightarrow v = \sqrt{\frac{E_{spring}}{m}} \approx 354 \frac{m}{s}$

faster than speed of sound!

Zwei Kinder nach Tauziehen umgekommen

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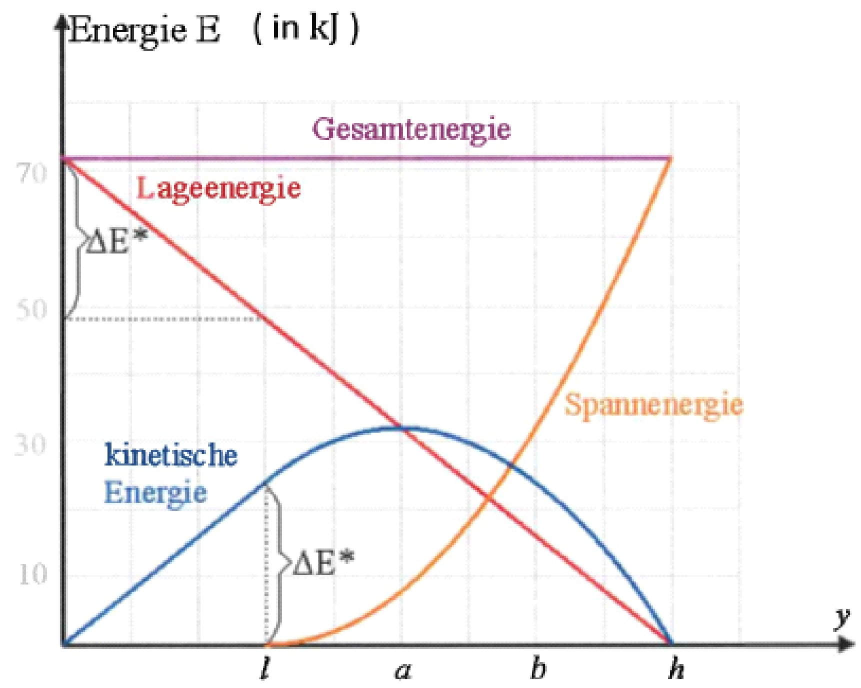


a) During the jump, what is happening at the depths indicated by $y = l$ and $y = a$ or why are those depths special? /1

For the following two subexercises, read off the required data from the diagram as accurately as possible.

b) How does the curve of total energy (*Gesamtenergie*) relate to the three other curves? Confirm your assumption numerically for height b. /1.5

m

c) What is the maximum speed reached by a jumper weighing 60 kg? /2

a) $y = l$: rope starts to tension (just free fall before) so l is the length of the relaxed rope.

$y = a$: highest kinetic energy $\Rightarrow$ highest speed

b) at all times: $E_{pot} + E_{kin} + E_{spring} = E_{total} \approx 72\,kJ$

$y = b$: $16\,kJ + 24\,kJ + 32\,kJ = 72\,kJ$ ✓

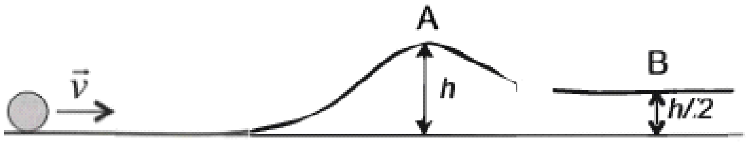
c) $y = a$: $E_{kin} \approx 32\,kJ = \frac{1}{2} m \cdot v^2$

$\Rightarrow v = \sqrt{\frac{2E_{kin}}{m}} \approx 32.7\,\frac{m}{s}$

5. Mini-Golf

4\

The picture shows a mini-golf ball with given mass m that moves with the initial speed $|\vec{v}|$ towards an obstacle. For simplicity, consider the motion to be *frictionless*.



a) When the ball arrives at point A, its speed $|\vec{v}_A|$ is almost zero. Derive a formula for the initial speed $|\vec{v}|$ depending on h. /1

b) Plug in the numbers $m = 15$ g, $h = 30$ cm to obtain the numeric result for $|\vec{v}|$. /1

c) Which percentage of the initial speed does the ball have when it arrives at point B? /2

a) $\frac{1}{2} m \cdot v^2 = m \cdot g \cdot h \Rightarrow v = \sqrt{2 \cdot g \cdot h}$

b) $v = \sqrt{2 \cdot g \cdot h} = \sqrt{2 \cdot 10 \frac{m}{s^2} \cdot 0.3 m} \approx 2.45 \frac{m}{s}$

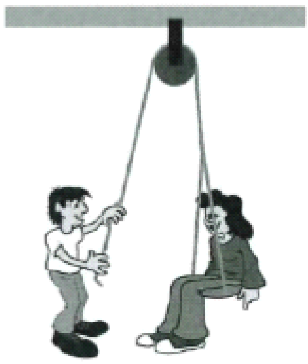
c) half of the potential energy at A has been converted to kinetic energy: $v_B = \sqrt{2 \cdot 10 \frac{m}{s^2} \cdot 0.15 m} \approx 1.73 \frac{m}{s}$

$\frac{v_B}{v} = \frac{1.73}{2.45} \approx 71\% \quad \left(= \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}\right)$

6. Dr Hansjakobli und ds Babettli

4\

Babettli (40 kg) sits on a board to which a rope is attached, that runs over a pulley at the ceiling.



a) What force does Hansjakobli have to exert to pull Babettli up? What is the minimum force that the suspension on the ceiling must be able to hold? /2

b) Now Hansjakobli hands Babettli the end of the rope. Does this change the situation? Can she pull herself up and if so, what force does she need to exert? What is the minimum force that the suspension on the ceiling must be able to hold now? /2

$L = m \cdot g = 400 N$

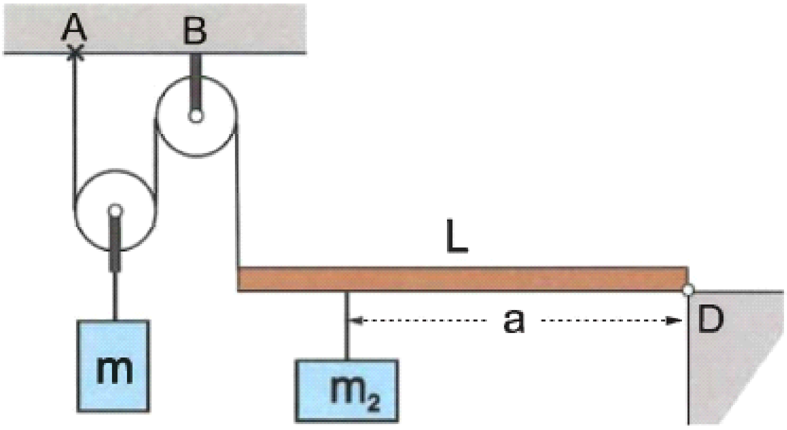
a) $n = 1: \quad F = \frac{L}{1} = 400 N$

$F_S = F \cdot 2 = 800 N$

b) $n = 2: \quad F = \frac{L}{2} = 200 N$

$F_S = F \cdot 2 = 400 N = L$

7. Pulley and Lever Combination 5\



Given: $L = 1\text{ m}$, $a = 80\text{ cm}$, $m_2 = 300\text{ g}$, mass of the pulleys 100 g each.
The lever can be considered massless.

a) Calculate the force with which the lever pulls on the rope. /1
b) Calculate the mass m needed for equilibrium. /2
c) Calculate the forces acting at the points A and B at the ceiling. /2

a) $F \cdot L = m_2 \cdot g \cdot a$

$$F = m_2 \cdot g \cdot \frac{a}{L} = 2.4\,N$$

b) $n = 2$: $F_G = 2 \cdot F = 4.8\,N$

$$m_{tot} = \frac{F_G}{g} = 0.48\text{ kg} = 480\text{ g}$$

$$m = m_{tot} - 100\text{ g} = 380\text{ g}$$

c) $F_A = F = 2.4\,N$

$$F_B = 2 \cdot F + 0.1\text{ kg} \cdot g = 4.8\,N + 1\,N = 5.8\,N$$